Stage 1 — Understanding Questions: Data Preparation and Epidemic Growth

Q1. Building the mpox weeks

Equation (1) constructs weekly incidence by differencing cumulative case counts on successive Mondays:

$$y_k = C_{OWID}(m_k) - C_{OWID}(m_{k-1}).$$

Week	Monday	Weekly count
0	9 May 2022	Reference Monday
1	16 May 2022	18
2	23 May 2022	15

Week 1: $y_1 = C(16\ \text{May}) - C(9\ \text{May}) = 18$ cases.

Week 2: $y_2 = C(23\ \text{May}) - C(16\ \text{May}) = 15$ cases.

These are the expected counts specified for mpox_us_weekly.csv. The raw daily CSV and processed CSV were not provided, so the individual cumulative totals and daily rows cannot be independently checked. As Section 2.1 explains, six zero entries followed by a Monday batch are a reporting convention, not observed daily infection incidence. Fitting such values as daily cases would confuse reporting schedules with epidemic dynamics.

Q2. Isotonic regression for Sierra Leone

Reported cumulative counts, in chronological order, are 5,235 (27 Oct), 5,338 (29 Oct), 4,759 (2 Nov), and 4,862 (4 Nov 2014). Isotonic regression minimizes the sum of squared deviations subject to nondecreasing fitted counts.

Minimize $\Sigma_{i=1}^{4} (y_i - c_i)^2$ subject to $c_1 \le c_2 \le c_3 \le c_4$.

Pooling step 1. Since 5,338 > 4,759, average the second and third reports:

$(5{,}338 + 4{,}759)/2 = 10{,}097/2 = 5{,}048.5.$

Provisional sequence: (5,235; 5,048.5; 5,048.5; 4,862).

Pooling step 2. Since 5,235 > 5,048.5, merge the first observation with the pooled block:

$(5{,}235 + 5{,}338 + 4{,}759)/3 = 15{,}332/3 = 5{,}110.6667.$

Provisional sequence: (5,110.6667; 5,110.6667; 5,110.6667; 4,862).

Pooling step 3. Since 5,110.6667 > 4,862, merge all four reports:

$(5{,}235 + 5{,}338 + 4{,}759 + 4{,}862)/4 = 20{,}194/4 = 5{,}048.5.$

Cleaned values for this four-report block: (5,048.5; 5,048.5; 5,048.5; 5,048.5). The full-series fit should be compared with ebola_reports.csv, which was not supplied: observations outside this four-report block

could change the final pooled block.

Deleting the 4,759 observation would discard genuine surveillance information without proof that this particular report was erroneous. Isotonic regression treats the sequence consistently, rather than requiring an arbitrary judgment about which individual report is wrong.

Q3. The take-off rule

The indexed series is: week 0 = 4; weeks 1–4 = 0,0,0,0; week 5 = 2; week 6 = 0; week 7 = 9; week 8 = 30; week 9 = 80; week 10 = 60.

Definition 1 requires at least three consecutive weeks with counts below 1. Weeks 1–4 form a qualifying quiet run of length four; week 6 is an isolated zero and does not qualify. The peak is 80 at week 9. The last qualifying quiet run before the peak ends at week 4.

> Take-off week: $k_0 = 4 + 1 = 5$.

Fitting from week 0 includes an early cluster followed by a prolonged flat stretch. A fractional model may absorb that reporting or epidemic lull by fitting an artificially small fractional order α, making memory appear stronger than it really is.

Q4. The spike flag

Weekly counts from 14 December 2025 to 18 January 2026: 18, 12, 174, 63, 165, 142. Definition 2 flags a week if its count exceeds three times the larger of the local-neighbor median and 5.

> Flag when $y_k > 3 \times \max\{\text{median(neighbors at } k-2, k-1, k+1, k+2), 5\}$.

28 December (174 cases). The four neighbors are 18, 12, 63, 165; sorted: 12, 18, 63, 165.

> Median = (18 + 63)/2 = 40.5; threshold = 3 × max(40.5, 5) = 121.5.
>
> 174 > 121.5, so 28 December IS FLAGGED.

11 January (165 cases). The supplied window provides three neighbors: 174, 63, 142. The fourth neighbor (25 January) is missing from the question.

> Available-neighbor median = median(63, 142, 174) = 142; threshold = 3 × 142 = 426.
>
> 165 < 426, so 11 January IS NOT FLAGGED using available neighbors.

Even with the missing fourth neighbor, the median of the four neighbors cannot be less than (63 + 142)/2 = 102.5, so the full-data threshold cannot be below 307.5. Hence the not-flagged decision is robust, although the exact full-data median cannot be determined without 25 January. The two weeks differ because the rule compares each count with its local context, not with a fixed absolute cutoff.

Q5. Reading the growth-shape figure

Exponential growth. Starting from $C(t) = C_0 e^{rt}$, take natural logarithms and apply the logarithm product rule:

> $\ln C(t) = \ln(C_0 e^{rt}) = \ln C_0 + \ln(e^{rt}) = \ln C_0 + rt$.

Thus ln C plotted against ordinary time t is a straight line with slope r and intercept $\ln C_0$.

Power-law growth. Starting from $C(t) = At^p$, for $A > 0$ and $t > 0$:

$$\ln C(t) = \ln(At^p) = \ln A + \ln(t^p) = \ln A + p \ln t.$$

Thus ln C plotted against ln t is a straight line with slope p and intercept ln A.

In Figure 3(a), Guinea and Sierra Leone have roughly straight sections over restricted intervals, but every trajectory bends downward and no complete series is a single straight line. In Figure 3(b), mpox and Liberia show relatively extended approximately straight rising portions; none of the full trajectories is perfectly linear, and the measles outbreaks are particularly irregular.

Two alternative causes of downward curvature: (1) Susceptible depletion reduces the number of people available for infection, slowing transmission. (2) Behavioral change or interventions can reduce the transmission rate over time. Both mechanisms operate in integer-order models and can create sub-exponential growth without genuine memory.

Therefore, the observed curvature is consistent with memory but is not evidence sufficient to identify it. Competing mechanistic fits, identifiability analysis, and out-of-sample evaluation are needed to distinguish explanations.